

Formulating state estimation as a weighted least-squares problem

$$\min_{\theta_1, \theta_2, \dots, \theta_n} \sum_{(i,j) \in M} \left[P_{ij}^{\text{meas}} - B_{ij} (\theta_i - \theta_j) \right]^2$$

$$\theta_i = 0 \quad (\text{one only})$$

$$x = \theta$$

$$Z_{\text{meas}} = \left[P_{ij}^{\text{meas}} \right]_{(i,j) \in M}$$

$$\min_{x_2, \dots, x_n} = \sum_{i=1}^m \left(\frac{Z_i - f_i(x)}{\sigma_i^2} \right)^2$$

all except
slack bus 1

$$f(x) = [H]x$$

$$R = \begin{bmatrix} \sigma_1^2 & & \\ & \ddots & \\ & & \sigma_m^2 \end{bmatrix}$$

$$\min_x J(x) = [Z_{\text{meas}} - f(x)]^T [R^{-1}] [Z_{\text{meas}} - f(x)]$$

Formulating state estimation as a weighted least-squares problem

$$\begin{aligned}\nabla J(x^*) &= 0 \\ &= -2H^T R^{-1} z_{meas} + 2H^T R^{-1} H x^*\end{aligned}$$

$$x^* = [H^T R^{-1} H]^{-1} H^T R^{-1} z_{meas}$$

$$n=m \quad x^* = H^{-1} z_{meas}$$

Example

$$H = \begin{bmatrix} 5 & -5 \\ 2.5 & 0 \\ 0 & -4 \end{bmatrix}$$

$$B_{12}(\theta_1 - \theta_2) \Rightarrow f_{12}(x)$$

$$\theta_3 = 0$$

$$B_{13}(\theta_1 - \theta_3)$$

$$B_{32}(\theta_3 - \theta_2)$$

$$R = \begin{bmatrix} 10^{-4} & & \\ & 10^{-4} & \\ & & 10^{-4} \end{bmatrix} \quad \left(Z_{meas} - Hx \right)^2$$

$\Rightarrow$

σ^2

$$\theta_1^* = 0.0285 \text{ rad}$$

$$\theta_2^* = -0.0943 \text{ rad}$$